

(TR-104) PYTHON WITH AI/ML

Training Day 44 Report

Date: 12 March 2026

Topic Covered: Fine-Tuning LLaMA 3.2 for RAG-Based AI Assistant & Hallucination Analysis

The forty-fourth day of training focused on improving the RAG-based AI Assistant by creating a structured company dataset, fine-tuning LLaMA 3.2 models using LoRA adapters, analyzing hallucination issues, and optimizing the system for accurate context-aware response generation.

- Extracted and analyzed company data from PDF, TXT, and HTML files for dataset preparation.
- Created a structured TechCorp Q&A dataset containing question-answer pairs related to products, pricing, support, shipping, policies, and company information.
- Fine-tuned the LLaMA 3.2 1B model on Google Colab T4 GPU using Unsloth and LoRA adapters.
- Identified hallucination issues in the 1B model where incorrect products and false company details were generated.
- Diagnosed the root cause and understood that smaller models struggle to follow strict instructions in RAG pipelines.
- Upgraded the workflow from LLaMA 3.2 1B to LLaMA 3.2 3B model for improved instruction-following capability.
- Retrained the 3B model with optimized hyperparameters including learning rate, epochs, and LoRA rank configuration.
- Exported the fine-tuned model into GGUF format using q4_k_m quantization for optimized local inference.
- Created an Ollama Modelfile with strict inference parameters such as low temperature and repeat penalty to reduce hallucinated outputs.
- Registered the fine-tuned model as techcorp-bot in Ollama for local inference and testing.
- Updated rag_pipeline.py with a strict context-only system prompt and repetition-removal functionality.

- Verified ChromaDB vector retrieval results using multiple test queries and validated relevant chunk retrieval.
- Updated the dataset multiple times to fix incorrect answers, remove hallucinated product information, and improve response consistency.
- Expanded the dataset with multiple wording variations to reduce hallucination caused by unseen query patterns.
- Analyzed the limitations of fine-tuned models when handling unseen semantic variations inside RAG systems.
- Strengthened understanding of fine-tuning, retrieval grounding, prompt engineering, and hallucination reduction techniques in production-level AI systems.
- GitHub Link:
<https://github.com/JaspinderKaurWalia26/finetuning/blob/main/finetuning.py>
- Blog Link:
<https://medium.com/@jaspinderkaurwalia855/building-a-hallucination-free-rag-chatbot-with-fine-tuned-llama-3-2-79e058d80689>